

Stable finite-type components and one-sided removal of a rank hypothesis

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Status. Candidate partial result, likely valid, awaiting human review. This does not completely remove the dimension hypothesis in Proposition 2.1 of de Laat–Mimura–de la Salle [1].

1 Source question

Let K be a compact Lie group with a left-invariant Riemannian metric d , let $U \subset K \times K$ be closed, and let U act on K by

$$(k', k'') \cdot k = k' k (k'')^{-1}.$$

For a finite-dimensional unitary representation $\rho : U \rightarrow \mathcal{U}(V)$, let U_k be the stabilizer of k , let

$$V_k = V^{\rho(U_k)}, \quad P_k : V \rightarrow V_k$$

be the invariant subspace and its orthogonal projection. Let λ denote the left regular representation of K on $L^2(K)$.

For a Banach space X , define

$$\Delta_X(k, k_0) = \left\| \left(\int_U (\lambda(u \cdot k) - \lambda(u \cdot k_0)) du \right) \otimes \text{Id}_X \right\|_{B(L^2(K; X))}. \quad (1)$$

An admissible finite-type coefficient is a map

$$f(a) = \sum_{i=1}^n \langle a \xi_i, \eta_i \rangle v_i, \quad \sum_i \|\xi_i\| \|\eta_i\| \|v_i\| \leq 1,$$

satisfying

$$f(\pi(k') a \pi((k'')^{-1})) = \rho(k', k'') f(a) \quad ((k', k'') \in U)$$

for an isometric representation $\pi : K \rightarrow O(X)$.

Proposition 2.1 of [1] proves

$$\|f(\pi(k)) - f(\pi(k_0))\| \leq C_{V, k_0} (d(k, k_0) + \Delta_X(k, k_0)) \quad (2)$$

under the hypothesis $\dim V_k = \dim V_{k_0}$. Immediately afterward the authors ask whether this dimension hypothesis is necessary. Figure 1 reproduces the complete source page.

The contragredient representation ${}^t\pi$ of a representation π of G on X is the representation of G on X^* given by $g \mapsto \pi(g^{-1})^*$. It might not be strongly continuous, even not if π is.

2.4. From estimates on invariant coefficients to estimates on finite type coefficients

In our proof of Theorem A, we will use the following result, which was proved in [29, Proposition 2.8] under the assumption that U is abelian. The setting is the following. Let K be a compact Lie group with a left-invariant Riemannian metric d , let $U \subset K \times K$ be a closed subgroup. Let λ be the left regular representation of K on $L^2(K)$. We denote by $u \cdot k$ the action of an element $u = (k', k'') \in U$ on an element $k \in K$ by left-right multiplication. For $k \in K$ we denote by $U_k \subset U$ the stabilizer of k for the action of U on K . If ρ is a finite-dimensional unitary representation of U on a Hilbert space V , we denote by $V_k \subset V$ the space of $\rho(U_k)$ -invariant vectors in V , and by $P_k \in B(V)$ the orthogonal projection onto V_k .

PROPOSITION 2.1. — *For every $k_0 \in K$ and every finite-dimensional unitary representation ρ of U on V , there exists a constant $C_{V, k_0} > 0$ such that the following holds: for every Banach space X , every isometric representation $\pi: K \rightarrow O(X)$ and every map $f: B(X) \rightarrow V$ of the form $f(a) = \sum_{i=1}^n \langle a\xi_i, \eta_i \rangle v_i$ satisfying*

- $\xi_i \in X$, $\eta_i \in X^*$ and $v_i \in V$,
- $\sum_{i=1}^n \|\xi_i\|_X \|\eta_i\|_{X^*} \|v_i\|_V \leq 1$,
- $f(\pi(k')a\pi(k''^{-1})) = \rho(k', k'')f(a)$ for all $a \in B(X)$ and $(k', k'') \in U$,

we have

$$\|f(\pi(k)) - f(\pi(k_0))\|_V \leq C_{V, k_0} \left(d(k, k_0) + \left\| \left(\int_U (\lambda(u \cdot k) - \lambda(u \cdot k_0)) du \right) \otimes \text{Id}_X \right\|_{B(L^2(K; X))} \right)$$

for all $k \in K$ such that $\dim(V_k) = \dim(V_{k_0})$.

We do not know whether the assumption $\dim(V_k) = \dim(V_{k_0})$ is necessary. At least in the case of $K = \text{SO}(n)$ and $U = \text{SO}(n-1) \times \text{SO}(n-1)$ (Lemma 4.2) of the case of [15] the proposition holds without this assumption, because the norm in $B(L^2(K))$ of $\int_U (\lambda(u \cdot k) - \lambda(u \cdot k_0)) du$ is greater than 1 otherwise.

We will use the following lemma.

Figure 1: Proposition 2.1 and the open question in [1, p. 1867, Section 2.4].

2 Idea of the proof

The source proof constructs a smooth covariant matrix coefficient $\psi : K \rightarrow B(V)$ with $\psi(k_0) = P_{k_0}$. Covariance forces $\psi(k)$ to vanish on $V_k^\perp$. Thus, near k_0 , the large singular-value subspace of $\psi(k)$ has the old rank $\dim V_{k_0}$ and lies inside V_k . This is a canonical “stable” part $Q_k V$ of the possibly larger invariant space V_k .

The source argument already controls $\psi(k)f(\pi(k)) - f(\pi(k_0))$. On the stable singular subspace, $\psi(k)$ has a uniformly bounded inverse. Applying this inverse recovers an estimate for $Q_k f(\pi(k)) - f(\pi(k_0))$. If $\dim V_k \leq \dim V_{k_0}$, the stable subspace exhausts V_k , so the original conclusion follows. The only directions not treated are invariant directions that appear when the stabilizer rank jumps upward.

3 Formal result

Put $d_0 = \dim V_{k_0}$. We use the following two facts proved in Section 2.4 of [1]. There is a Lipschitz map $\psi : K \rightarrow B(V)$ such that

$$\psi(u \cdot k) = \psi(k)\rho(u^{-1}), \quad \psi(k_0) = P_{k_0}, \quad (3)$$

and its scalar entries are coefficients of the regular representation. Moreover, the reduction to invariant coefficients in the proof of Proposition 2.1 gives, for every admissible X, π, f ,

$$\|\psi(k)f(\pi(k)) - f(\pi(k_0))\| \leq C_0 \Delta_X(k, k_0), \quad (4)$$

where C_0 depends only on V, k_0 and the chosen ψ .

Lemma 1 (Stable spectral subspace). *There are $r > 0$ and $C_1 < \infty$ such that, whenever $d(k, k_0) < r$, there is an orthogonal projection Q_k satisfying*

$$Q_k \leq P_k, \quad \text{rank } Q_k = d_0, \quad \|Q_k - P_{k_0}\| \leq C_1 d(k, k_0).$$

There is also an operator $B_k \in B(V)$ such that

$$B_k \psi(k) Q_k = Q_k, \quad \sup_{d(k, k_0) < r} \|B_k\| < \infty, \quad \|B_k - P_{k_0}\| \leq C_1 d(k, k_0).$$

Finally,

$$\|\psi(k)(I - Q_k)\| \leq C_1 d(k, k_0).$$

Proof. Write $P_0 = P_{k_0}$ and $A_k = \psi(k)^* \psi(k)$. From (3), if $u \in U_k$ then $\psi(k) = \psi(k)\rho(u^{-1})$. Averaging over U_k gives

$$\psi(k) = \psi(k)P_k, \quad (5)$$

so $A_k = P_k A_k P_k$.

Since ψ is Lipschitz and $\psi(k_0) = P_0$, the positive matrices A_k satisfy

$$\|A_k - P_0\| \leq C d(k, k_0).$$

Choose r so small that the spectrum of A_k lies in

$$[0, 1/4] \cup [3/4, 5/4]$$

for $d(k, k_0) < r$, with exactly d_0 eigenvalues in the second interval. Define

$$Q_k = \mathbf{1}_{[1/2, \infty)}(A_k).$$

Finite-dimensional spectral calculus, with the spectral gap fixed, gives

$$\|Q_k - P_0\| \leq C\|A_k - P_0\| \leq C_1 d(k, k_0).$$

Equation (5) implies $Q_k \leq P_k$.

On $Q_k V$, the operator A_k is bounded below by $3/4$. Set

$$B_k = Q_k(A_k|_{Q_k V})^{-1} Q_k \psi(k)^*.$$

Then $B_k \psi(k) Q_k = Q_k$. Inversion and spectral projection are Lipschitz on this fixed gapped neighborhood of P_0 , so B_k is uniformly bounded and

$$\|B_k - P_0\| \leq C_1 d(k, k_0).$$

Lastly,

$$\begin{aligned} \|\psi(k)(I - Q_k)\| &\leq \|(\psi(k) - P_0)(I - Q_k)\| + \|P_0(I - Q_k)\| \\ &\leq \|\psi(k) - P_0\| + \|P_0 - Q_k\| \leq C_1 d(k, k_0). \end{aligned}$$

Enlarging C_1 yields all assertions. $\square$

Theorem 2 (The stable component is controlled). *With no hypothesis on $\dim V_k$, for every k satisfying $d(k, k_0) < r$ and every admissible X, π, f ,*

$$\|Q_k f(\pi(k)) - f(\pi(k_0))\| \leq C(d(k, k_0) + \Delta_X(k, k_0)). \quad (6)$$

Proof. Put $v = f(\pi(k))$ and $v_0 = f(\pi(k_0))$. Admissibility and the fact that $\pi(k)$ is an isometry give $\|v\|, \|v_0\| \leq 1$. Equivariance of f gives $v \in V_k$ and $v_0 \in V_{k_0}$, hence $P_0 v_0 = v_0$.

Using Lemma 1,

$$\begin{aligned} Q_k v - v_0 &= B_k \psi(k) Q_k v - P_0 v_0 \\ &= B_k(\psi(k)v - v_0) + (B_k - P_0)v_0 - B_k \psi(k)(I - Q_k)v. \end{aligned}$$

The first term is bounded by a constant times $\Delta_X(k, k_0)$ by (4); the other two are bounded by a constant times $d(k, k_0)$ by Lemma 1. This proves (6). $\square$

Corollary 3 (One-sided removal of the rank equality). *The conclusion (2) of Proposition 2.1 in [1] holds for every $k \in K$ satisfying*

$$\dim V_k \leq \dim V_{k_0}.$$

Thus the source hypothesis $\dim V_k = \dim V_{k_0}$ may be replaced by the one-sided inequality above.

Proof. If $d(k, k_0) < r$, Lemma 1 gives

$$d_0 = \text{rank } Q_k \leq \dim V_k \leq d_0.$$

Hence $Q_k = P_k$. Since $f(\pi(k)) \in V_k$, Theorem 2 becomes (2).

If $d(k, k_0) \geq r$, admissibility gives the trivial estimate

$$\|f(\pi(k)) - f(\pi(k_0))\| \leq 2 \leq \frac{2}{r} d(k, k_0).$$

Combining the two regions proves the claim with a constant depending only on V, k_0 . $\square$

Corollary 4 (Necessary shape of a counterexample). *If the completely rank-free version of Proposition 2.1 is false, then there is a sequence $k_j \rightarrow k_0$ such that*

$$\dim V_{k_j} > \dim V_{k_0}, \quad \left\| \int_U (\lambda(u \cdot k_j) - \lambda(u \cdot k_0)) du \right\|_{B(L^2(K))} \rightarrow 0.$$

For such a sequence, the only component not controlled by Theorem 2 is

$$(P_{k_j} - Q_{k_j})f(\pi(k_j)),$$

which lies in the newly appearing stabilizer-invariant directions.

Proof. Failure of a uniform constant in (2), together with the bound of 2 on its left side, produces a sequence for which the right-hand denominator tends to zero. Thus $k_j \rightarrow k_0$. For every nonzero X , restricting the amplified operator in (1) to scalar functions tensored with one fixed unit vector of X shows that its norm dominates the displayed scalar norm. Hence that scalar norm also tends to zero. Corollary 3 excludes $\dim V_{k_j} \leq d_0$. The component statement follows from $Q_{k_j} \leq P_{k_j}$ and Theorem 2. $\square$

4 Verification and limitations

The new proof is finite-dimensional and contains no numerical dependency. Its two points requiring expert verification are: (i) $A_k = P_k A_k P_k$, which ensures $Q_k \leq P_k$; and (ii) the Moore–Penrose inverse identity in Theorem 2. Both are explicit in the displayed calculation.

This packet is a partial answer only. It does not bound the extra-rank component $(P_k - Q_k)f(\pi(k))$. A full positive answer would follow from a uniform lower bound on the scalar orbital norm whenever such new invariant directions occur, but no general proof of that assertion is supplied.

A bounded web/arXiv search on 21 July 2026 used the exact source sentence and close variants of the phrases “dimension of V_k ,” “finite-type coefficients,” “weak relative property (T_X),” and “strong property (T).” It found the source paper and related strong-property-(T) work, but no later settlement of this exact rank question. The novelty assessment is therefore provisional and not an exhaustive citation review.

Human-review recommendation. Check Lemma 1 and the three-term identity in the proof of Theorem 2. If confirmed, accept Corollary 3 and the stable-component theorem as a partial result; do not treat the original rank-free question as solved.

References

- [1] T. de Laat, M. Mimura, and M. de la Salle, *On strong property (T) and fixed point properties for Lie groups*, Ann. Inst. Fourier (Grenoble) **66** (2016), no. 5, 1859–1893; arXiv:1508.05860.